

# ADR Cards

One card per decision. Each card shows the number, scope, context, status, the decision in one breath, and why it was taken. Generated from the ADR front matter; the full record is the linked file under `obsidian/V3-Platform/Architecture/ADR/`.

| ADR     | SCOPE · CONTEXT      | TITLE                                                                           | STATUS   | DATE       |
|---------|----------------------|---------------------------------------------------------------------------------|----------|------------|
| ADR-046 | LAB<br>ORGANIZATIONS | Transporter Aggregate Boundary: Shared Identity, Separate Vetting               | Accepted | 2026-08-20 |
| ADR-047 | LAB<br>ORGANIZATIONS | Temporal Saga Design for TransporterProfile Vetting                             | Accepted | 2026-08-20 |
| ADR-048 | LAB<br>ORGANIZATIONS | Document Blob Storage: NATS Object Store                                        | Accepted | 2026-08-20 |
| ADR-049 | LAB<br>ORGANIZATIONS | Cross-Aggregate Concurrency: Two Mechanisms, One Guard                          | Accepted | 2026-08-20 |
| ADR-050 | LAB<br>ORGANIZATIONS | GIT Certificate Change Log Needs a Provenance Source                            | Accepted | 2026-08-22 |
| ADR-051 | LAB<br>ORGANIZATIONS | Entity Identity Is a ULID, Not a UUID                                           | Accepted | 2026-08-24 |
| ADR-052 | LAB DATA             | One Postgres Instance, One Database and One Role per Service                    | Accepted | 2026-09-03 |
| ADR-053 | V3 DATA              | Use Shared PostgreSQL Instances by Default                                      | Accepted | 2026-09-03 |
| ADR-054 | V3 PLATFORM          | Portability Rules for Multi-Region Deployment and Integration                   | Accepted | 2026-09-07 |
| ADR-055 | LAB PLATFORM         | Split the Lab Compose File into Cell Tiers, and Make Every Host Port a Variable | Accepted | 2026-09-08 |

■ lab — Tech Lab / Dictionary POC decision, implemented in this repo

■ v3 — Proposed Linebooker V3 platform principle

■ Accepted

■ Proposed

046

LAB · ORGANIZATIONS

# Transporter Aggregate Boundary: Shared Identity, Separate Vetting

**ACCEPTED** 2026-08-20 ADR-046-lab-organizations-transporter-aggregate-split.md

DECISION

Organization stays the single identity aggregate for Shipper and Transporter. Vetting state lives in a new event-sourced TransporterProfile aggregate.

WHY

Identity is current-state CRUD. Vetting has a lifecycle that must replay and audit. One aggregate would force both onto one persistence model.

RELATIONSHIPS

related ADR-047, ADR-048, ADR-049, ADR-050, ADR-051

LINKED DECISIONS

- ADR-047** Temporal Saga Design for TransporterProfile Vetting
- ADR-048** Document Blob Storage: NATS Object Store
- ADR-049** Cross-Aggregate Concurrency: Two Mechanisms, One Guard
- ADR-050** GIT Certificate Change Log Needs a Provenance Source
- ADR-051** Entity Identity Is a ULID, Not a UUID

047

LAB · ORGANIZATIONS

# Temporal Saga Design for TransporterProfile Vetting

ACCEPTED

2026-08-20   ADR-047-lab-organizations-transporter-vetting-temporal-saga.md

DECISION

Vetting runs as a Temporal workflow with a two-branch parallel saga. Activities publish the domain events. Compensation is forward-only.

WHY

A multi-day, human-in-the-loop approval with real compensation needs durable timers, signals and saga support. That is what Temporal is for.

RELATIONSHIPS

related ADR-046, ADR-048, ADR-049, ADR-050

LINKED DECISIONS

ADR-046

 Transporter Aggregate Boundary: Shared Identity, Separate Vetting

ADR-048

 Document Blob Storage: NATS Object Store

ADR-049

 Cross-Aggregate Concurrency: Two Mechanisms, One Guard

ADR-050

 GIT Certificate Change Log Needs a Provenance Source

048

LAB · ORGANIZATIONS

Document Blob Storage: NATS Object Store

ACCEPTED

2026-08-20    ADR-048-lab-organizations-document-storage-nats-object-store.md

DECISION

Compliance document bytes go in a NATS Object Store bucket per tenant account, not S3, MinIO or GCS.

WHY

The lab evaluates NATS patterns and tenant isolation already comes from the NATS account. The cost is real limits on size and durability, and the ADR records them.

RELATIONSHIPS

related ADR-046, ADR-047, ADR-050

LINKED DECISIONS

- ADR-046   Transporter Aggregate Boundary: Shared Identity, Separate Vetting
- ADR-047   Temporal Saga Design for TransporterProfile Vetting
- ADR-050   GIT Certificate Change Log Needs a Provenance Source

049

LAB · ORGANIZATIONS

# Cross-Aggregate Concurrency: Two Mechanisms, One Guard

ACCEPTED

2026-08-20    ADR-049-lab-organizations-cross-aggregate-concurrency.md

DECISION

Organization uses optimistic row versioning. TransporterProfile uses per-subject expected-sequence on the TRANSPORTER stream. One guard sits across both.

WHY

The two aggregates persist differently, so no single mechanism covers both. The cross-aggregate invariant is a one-time gate, not a running constraint.

RELATIONSHIPS

related ADR-046, ADR-047, ADR-051

LINKED DECISIONS

- ADR-046**   Transporter Aggregate Boundary: Shared Identity, Separate Vetting
- ADR-047**   Temporal Saga Design for TransporterProfile Vetting
- ADR-051**   Entity Identity Is a ULID, Not a UUID

050

LAB · ORGANIZATIONS

# GIT Certificate Change Log Needs a Provenance Source

ACCEPTED

2026-08-22

ADR-050-lab-organizations-git-certificate-change-log-provenance.md

DECISION

The GOODS\_IN\_TRANSIT document is event-sourced onto the TRANSPORTER stream. Its Postgres row becomes a projection. The other four document types stay CRUD.

WHY

A change log needs an event history that did not exist. Only the GIT certificate has a business need for one, so only it pays the event-sourcing cost.

RELATIONSHIPS

related ADR-046, ADR-047, ADR-048

LINKED DECISIONS

**ADR-046** Transporter Aggregate Boundary: Shared Identity, Separate Vetting

**ADR-047** Temporal Saga Design for TransporterProfile Vetting

**ADR-048** Document Blob Storage: NATS Object Store

051

LAB · ORGANIZATIONS

Entity Identity Is a ULID, Not a UUID

ACCEPTED

2026-08-24   ADR-051-lab-organizations-ulid-entity-identity.md

DECISION

organizations-service mints 26-character ULIDs in Go. ID columns are TEXT with no default. shipping-service and accounts-service stay on UUID.

WHY

IDs sit in NATS subject tokens and KV keys, so they must be subject-safe and immutable. ULIDs sort by time and never contain a dot.

RELATIONSHIPS

related ADR-046, ADR-049

LINKED DECISIONS

ADR-046   Transporter Aggregate Boundary: Shared Identity, Separate Vetting

ADR-049   Cross-Aggregate Concurrency: Two Mechanisms, One Guard

052

LAB · DATA

# One Postgres Instance, One Database and One Role per Service

ACCEPTED

2026-09-03    ADR-052-lab-data-one-postgres-instance-database-per-service.md

DECISION

The six lab services share one Postgres container. Each owns one database and one role. CONNECT is revoked from PUBLIC and granted to the owner only.

WHY

Seven containers bought process isolation nothing needed. Per-role CONNECT privileges keep the real guarantee: a service cannot read another service's tables.

RELATIONSHIPS

applies ADR-053

LINKED DECISIONS

**ADR-053** Use Shared PostgreSQL Instances by Default

053

V3 · DATA

## Use Shared PostgreSQL Instances by Default

ACCEPTED

2026-09-03 ADR-053-v3-data-shared-postgres-by-default.md

### DECISION

Share PostgreSQL infrastructure by default within a region. Give every service its own logically isolated database, credentials and migrations. Move a database to dedicated infrastructure only on a demonstrated requirement.

### WHY

Dedicated instances per database cost more, fragment capacity and multiply operational work. Logical isolation by database, role and privileges keeps service ownership strong without that cost.

### RELATIONSHIPS

applied by ADR-052

### LINKED DECISIONS

**ADR-052** One Postgres Instance, One Database and One Role per Service

054

V3 · PLATFORM

## Portability Rules for Multi-Region Deployment and Integration

ACCEPTED

2026-09-07 ADR-054-v3-platform-portability-rules-multi-region.md

### DECISION

Every managed or vendor-specific dependency sits behind a port with at least two working adapters, a region runs on plain compute, disk and network, and no global component holds a copy of regional data. Ten numbered rules govern multi-region deployment, clustering and integration, including which of three cross-border shapes a residency or sovereignty requirement selects.

### WHY

Linebooker V3 must be liftable between clouds and into a customer's own datacentre. A lock-in decision is cheap to take and expensive to reverse, so the constraint has to be written down before the choices are made, not after.

### RELATIONSHIPS

related ADR-048, ADR-053

### LINKED DECISIONS

**ADR-048** Document Blob Storage: NATS Object Store

**ADR-053** Use Shared PostgreSQL Instances by Default

055

LAB · PLATFORM

## Split the Lab Compose File into Cell Tiers, and Make Every Host Port a Variable

ACCEPTED

2026-09-08 ADR-055-lab-platform-compose-split-cell-and-global.md

### DECISION

The one flat docker-compose.yml becomes four tier files under deploy/cell/ (infra, control, runtime, dedicated) plus an include-only compose.yaml. Every published host port is a \${VAR:-default} fed by one env file per cell, and container\_name is removed everywhere.

### WHY

A second local region could not start. Every host port and every container name was a literal, so two cells collided on both. Files split by when a thing ships; projects split by where it runs.

### RELATIONSHIPS

applies ADR-054  
related ADR-052

### LINKED DECISIONS

**ADR-052** One Postgres Instance, One Database and One Role per Service  
**ADR-054** Portability Rules for Multi-Region Deployment and Integration